Novelty after switching away: revised estimation specification

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Economic interpretation

Switching toward berry in the previous period can renew the appeal of plain this period. Novelty N is a binary, one-period recovery event for the flavor switched away from. It is not a first-ever purchase indicator, a continuous measure of unfamiliarity, or a boost for the flavor switched toward. The coefficient remains unrestricted: a positive coefficient supports the proposed recovery mechanism; a negative coefficient implies the opposite.

Definition and timing

Let w_{ijs} denote flavor j's share of purchased yogurt packages in calendar week s. Dominance means a strict majority, $w_{ijs} > 1/2$. For two consecutive observed purchase weeks $s-1, s$, define

$$\begin{aligned} E_{is} &= \mathbf{1}\{\exists k : w_{ik,s-1} = 0,\ w_{iks} > 0\}, \\ T_{ijs} &= \mathbf{1}\{w_{ij,s-1} > 1/2,\ \exists k \neq j : w_{iks} > 1/2\}, \\ A_{ijs} &= \mathbf{1}\{T_{ijs} = 1 \text{ or } (E_{is} = 1 \text{ and } w_{ijs} < w_{ij,s-1})\}. \end{aligned}$$

The revised predictor is $N_{ijt} = A_{ij,t-1}$, provided purchase baskets are observed in all three consecutive calendar weeks $t-2, t-1, t$; otherwise $N_{ijt} = 0$. Thus the first two observed weeks cannot receive the boost. A gap resets event eligibility, but does not erase accumulated stock.

A takeover rewards the displaced majority flavor. An introduction rewards only flavors whose shares decline. Multiple triggers still give $N = 1$, not multiple bonuses. A majority lost to a tie, without an introduction or another flavor gaining a majority, does not trigger novelty. No event means no boost, even if the previous event's basket composition persists.

Utility and unchanged stock

$$\begin{aligned} H_{ijt} &= \sum_{s<t,\ s \text{ observed}} w_{ijs}, \\ V_{ijt} &= \beta_0 + \beta_j + \alpha_i p_{ijt} + \gamma_i N_{ijt} + \phi_i H_{ijt}. \end{aligned}$$

The outside utility normalization and optional price-control residual are unchanged. The three random coefficients retain their current distributions. Current purchases update stock and the switching event only after current predictors are recorded. Histories are constructed before likelihood exclusions. The stock does not decay: recovery enters as a separate temporary utility term. A positive ϕ_i represents reinforcement; a negative one represents satiation. With binary N, γ_i is the utility difference when the event turns on, holding stock and other predictors fixed.

Examples

The table shows transitions from $t-2$ to $t-1$, followed by the boost in t. All shares are within yogurt purchases; all weeks must be consecutive.

Earlier basket	Later basket	Flavor receiving boost
Plain 2/3, berry 1/3	Plain 1/3, berry 2/3	Plain
All plain	Plain 2/3, berry 1/3	Plain
Plain 1/3, berry 2/3	Same composition	None
Plain 1/3, berry 2/3	All berry	None
All plain	All berry	Plain (once)

Comparison and server execution

The legacy predictor is $1 - w_{ij,t-1}$ when last week's purchase is observed. It remains available as `previous_share`; the new option is `lagged_switch_away`. The dedicated shell selects the new option and reestimates both controlled and uncontrolled heterogeneous-history demand using the prepared sample and numerical settings saved in an existing rebuild. No raw-data rebuild or pricing exercise is run. Earlier pricing dynamics have not been adapted to this new event rule.

Submit from the server checkout:

```
sbatch Code/final_simple/server/run_novelty_estimation.sh
/scratch/dtm63837/Kilts_Panel/rebuilds/run_JOB_ID
```

The submission is one command with the rebuild directory as its argument. Each job writes to a new subdirectory of that rebuild. All empirical I/O uses the existing research-server policy. Recover only the job's `RESULTS_ONLY` directory: estimates, covariance, aggregate distribution figures and summaries, first-stage estimates, and sanitized fit summaries. Weekly inputs and full internal fit records remain on the server.